

COMPREHENSIVE FINAL EXAMINATION

DevOps, Cloud Infrastructure, Kubernetes & Web Security

Date: May 13, 2026 | Duration: 3 Hours | Total Marks: 100

Instructions: This exam contains 15 Multiple Choice Questions (30 marks) and 10 Short Answer Questions (70 marks). Answer all questions. For MCQs, select the best answer. For short answers, provide concise and precise responses.

PART A: MULTIPLE CHOICE QUESTIONS (30 marks - 2 marks each)

1. In Kubernetes, what is the smallest deployable unit that can be managed?

- A) Container
- B) Pod
- C) Deployment
- D) ReplicaSet

2. Which Terraform command checks the state file and determines what changes are needed WITHOUT applying them?

- A) terraform validate
- B) terraform apply
- C) terraform plan
- D) terraform init

3. In SQL Injection attacks using the UNION keyword, why is NULL commonly used in the initial testing phase?

- A) It's faster to process
- B) It's type-flexible and can fit into any column
- C) It bypasses database validation
- D) It reduces server load

4. What does the 'create_before_destroy' lifecycle setting in Terraform accomplish?

- A) Deletes old resources before creating new ones
- B) Creates replacement resources first, then deletes old ones
- C) Prevents resource deletion
- D) Schedules resource updates

5. In a Kubernetes cluster, what is the role of ETCD?

- A) Schedule containers on nodes
- B) Execute containers
- C) Store cluster state and configuration data
- D) Manage network policies

6. Which type of Kubernetes Service exposes an application externally using a cloud provider's load balancer?

- A) ClusterIP
- B) NodePort
- C) LoadBalancer
- D) ExternalName

7. What is the primary difference between CI and CD in a CI/CD pipeline?

- A) CI deploys to production while CD tests locally
- B) CI automates testing and integration; CD automates deployment
- C) CI and CD are the same thing
- D) CI is for development; CD is for operations

8. In Web Application Security, what does SSRF stand for and what is its primary risk?

- A) Server-Side Resource Failure; causes application crashes
- B) Server-Side Request Forgery; allows attackers to make requests from the server
- C) Secure Server Response Format; relates to encryption
- D) Stateless Session Request Framework; relates to session handling

9. Which of the following is NOT a core DevOps principle?

- A) Automation of the software development lifecycle
- B) Collaboration and communication
- C) Manual testing before deployment
- D) Continuous improvement and minimization of waste

10. What is the purpose of a Kubernetes Liveness Probe?

- A) Determine if a pod is ready to receive traffic
- B) Determine if a pod should be restarted
- C) Monitor pod memory usage
- D) Control pod scheduling

11. In Terraform, what is the purpose of the 'terraform state' file?

- A) Define infrastructure configuration
- B) Track the current state of managed resources
- C) Store sensitive credentials
- D) Document best practices

12. Which OWASP vulnerability involves attackers uploading malicious files to a web server?

- A) Path Traversal
- B) Cross-Site Scripting
- C) File Upload Vulnerability
- D) SQL Injection

13. What is the relationship between containers and virtual machines?

- A) Containers replace VMs entirely
- B) Containers share OS kernel; VMs have separate OS instances
- C) VMs are lightweight; containers are heavy
- D) They are functionally identical

14. In Terraform, which data source allows you to retrieve information from AWS without creating resources?

- A) resource
- B) module
- C) data
- D) output

15. What does the 'restrictive access' principle suggest in cryptographic key management?

- A) Share keys with trusted parties only
- B) Encrypt all keys automatically
- C) Limit access to sensitive keys to authorized personnel only
- D) Store keys in plain text for accessibility

PART B: SHORT ANSWER QUESTIONS (70 marks - 7 marks each)

1. Explain the difference between Declarative and Imperative approaches in Infrastructure as Code (IaC). Provide an example of how each approach would handle updating a web server configuration.

[7 marks]

2. Describe the three pillars of CI/CD (Continuous Integration, Continuous Delivery, and Continuous Deployment). What problems does each pillar solve in the software development lifecycle?

[7 marks]

3. In Kubernetes, explain the concept of 'Readiness Probe' and 'Liveness Probe'. When would you use each one, and what happens when they fail?

[7 marks]

4. You are implementing a Terraform infrastructure for a production environment with a team of 5 developers. Why should you NOT store the terraform.tfstate file in Git version control? What are the risks and recommended alternatives?

[7 marks]

5. Explain how SQL Injection attacks work using the UNION keyword. Provide a step-by-step explanation of how an attacker would determine the number of columns in the original query.

[7 marks]

6. Describe the concept of 'Idempotency' in Infrastructure as Code. Why is this principle critical when managing cloud infrastructure, and what problems can arise if operations are not idempotent?

[7 marks]

7. In Kubernetes, explain the difference between a Deployment and a ReplicaSet. How does a Deployment manage rolling updates, and why is this approach better than directly managing ReplicaSets?

[7 marks]

8. Compare path traversal attacks with file upload vulnerabilities. How can a web application defend against both types of attacks? Provide specific mitigation strategies for each.

[7 marks]

9. Explain how TLS/SSL provides data encryption, data integrity, and authentication. Why are public-private key cryptography and digital certificates essential components of this process?

[7 marks]

10. In the context of Kubernetes Ingress Controllers, compare the two main types: web server-based (e.g., NGINX) and cloud load balancer-based. What are the advantages and disadvantages of each approach in terms of cost, complexity, and functionality?

[7 marks]

END OF EXAMINATION

Ensure all answers are clearly written. Show your working where applicable. Good luck!